



B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

EE 19-209-0406 ELECTROMAGNETIC FIELD THEORY (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Apply the basic concepts of coulomb's law and Gauss's law and analyze the boundary conditions.
 CO2: Calculate the force, electric field intensity and capacitance of different charge distributions.
 CO3: Calculate the inductance of co-axial cable and overhead transmission lines making use of boundary conditions for different media.
 CO4: Apply the basic knowledge of electromagnetic waves in real life situations such as wave propagation in transmission lines.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) State and prove Stoke's Theorem.	3	L2	1	1,2
(b) Derive Laplace and Poisson's Equations.	3	L1	1	1,2
(c) State the differential and integral form of Maxwell's equations for time varying fields.	3	L2	2	1,2
(d) Explain Ampere's Circuit Law. Give one application of Ampere's circuit law.	3	L3	2	1,3,6
(e) Write short note on skin depth and skin effect.	3	L2	3	1,3
(f) Discuss any one type of wave polarization.	3	L1	3	2,6
(g) What is the significance of Brewster angle?	3	L4	4	2,6
(h) Explain VSWR.	3	L2	4	2,6

PART B

(4 × 12 = 48)

II. (a) Transform the vector $A = 3\hat{a}_x + 4\hat{a}_y + 8\hat{a}_z$ at the point $p(x=7, y=8, z=9)$ into cylindrical co-ordinates system.	8	L4	1	1,3,6
(b) If $E = xy\hat{a}_x + x^2\hat{a}_y$, find	4	L1	1	2,3
(i) electric flux density, D.				
(ii) the volume charge density, ρ_v .				
OR				
III. (a) Derive the expression for capacitance of a coaxial cylinder.	8	L1	1	1,2
(b) State and explain Gauss's Law.	4	L2	1	3,6
IV. Obtain an expression for magnetic field intensity H for infinitely long co-axial cable using Ampere's circuital law.	12	L1	2	1,3,6
OR				
V. (a) Derive the expression for magnetic field intensity at a point along 'z' axis due to a circular loop lying in the xy plane using Biot-Savart's law.	8	L1	2	2,3
(b) Derive continuity equation.	4	L2	2	1,6

(P.T.O.)

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		Marks	BL	CO	PO
VI.	State and prove Poynting theorem and explain its significance.	12	L2	3	2,3
	OR				
VII.	Derive the solution of wave equation for lossy dielectric.	12	L1	3	1,6
VIII.	(a) Derive the wave equation for voltage and current for transmission line.	8	L2	4	1,2
	(b) Explain Snell's law.	4	L1	4	3,6
	OR				
IX.	Describe the expression for reflection and transmission coefficient for uniform plane wave incident normally on a perfect dielectric.	12	L3	4	3,6

Blooms's Taxonomy Levels

L1 - 38.33%, L2 - 40%, L3 - 12.5%, L4 - 9.2%.
